aquatic and terrestrial modes of life. They grow in shallow water. Examples: *Ranunculus*, *Typha* and *Sagittaria*.

Hygrophytes: The plants which can grow in moist damp and shady places are called hygrophytes. Examples: *Habenaria* (Orchid), Mosses (Bryophytes), etc.

Morphological adaptations of Hydrophytes:

In root

- Roots are totally absent in *Wolffia* and *Salvinia* or poorly developed in *Hydrilla* or well developed in *Ranunculus.*
- The root caps are replaced by **root pockets**. Example: *Eichhornia*

In stem

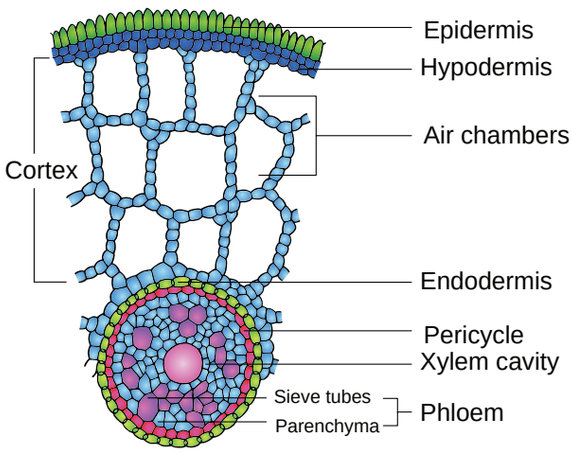
- The stem is long, slender, spongy and flexible in submerged forms.
- In free floating forms the stem is thick, short stoloniferous and spongy; and in rooted floating forms, it is a rhizome .
- Vegetative propagation is through runners, stolon, stem and root cuttings , tubers, dormant apices and offsets.

In leaves

- The leaves are thin, long and ribbon shaped in *Vallisneria* or long and linear in *Potamogeton* or finely dissected in *Ceratophyllum*
- The floating leaves are large and flat as in *Nymphaea* and *Nelumbo*. In *Eichhornia* and *Trapa* petioles become swollen and spongy.
- In emergent forms, the leaves show **heterophylly** (Submerged leaves are dissected and aerial leaves are entire). Example: *Ranunculus, Limnophila heterophylla* and *Sagittaria*

Anatomical adaptations

- Cuticle is either completely absent or if present it is thin and poorly developed
- Single layer of epidermis is present
- Cortex is well developed with aerenchyma
- Vascular tissues are poorly developed. In emergent forms vascular elements are well developed.
- Mechanical tissues are generally absent except in some emergent forms. Pith cells are sclerenchymatous.



T.s of *Hydrilla* stem

Figure 6.19: T.S. of *Hydrilla* stem

Physiological adaptations of Hydrophytes:

- Hydrophytes have the ability to withstand anaerobic conditions .
- They possess special aerating organs.

Xerophytes

The plants which are living in dry or xeric condition are known as **Xerophytes**. Xerophytic habitat can be of two different types. They are:

a. Physical dryness: In these habitats, soil has a little amount of water due to the inability of the soil to hold water because of low rainfall.

b. Physiological dryness: In these habitats, water is sufficiently present but plants are unable to absorb it because of the absence of capillary spaces. Example: Plants in salty and acidic soil.

Based on adaptive characters xerophytes are classified into three categories. They are Ephemerals, Succulents and Non succulent plants.

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